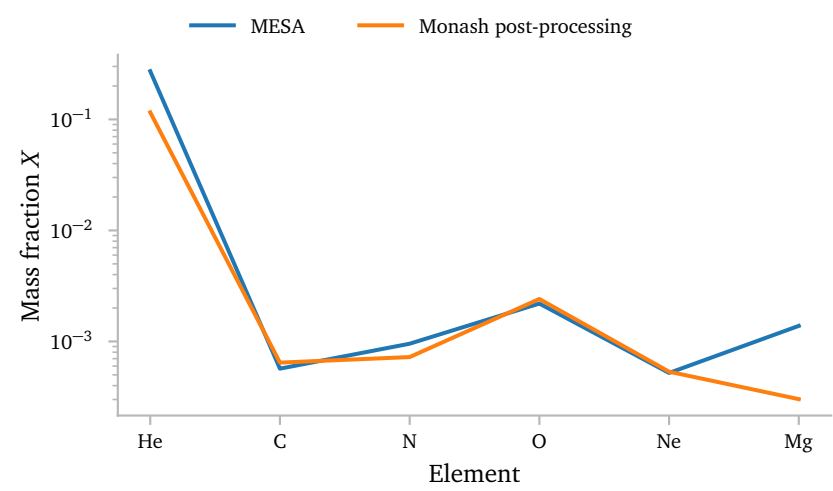


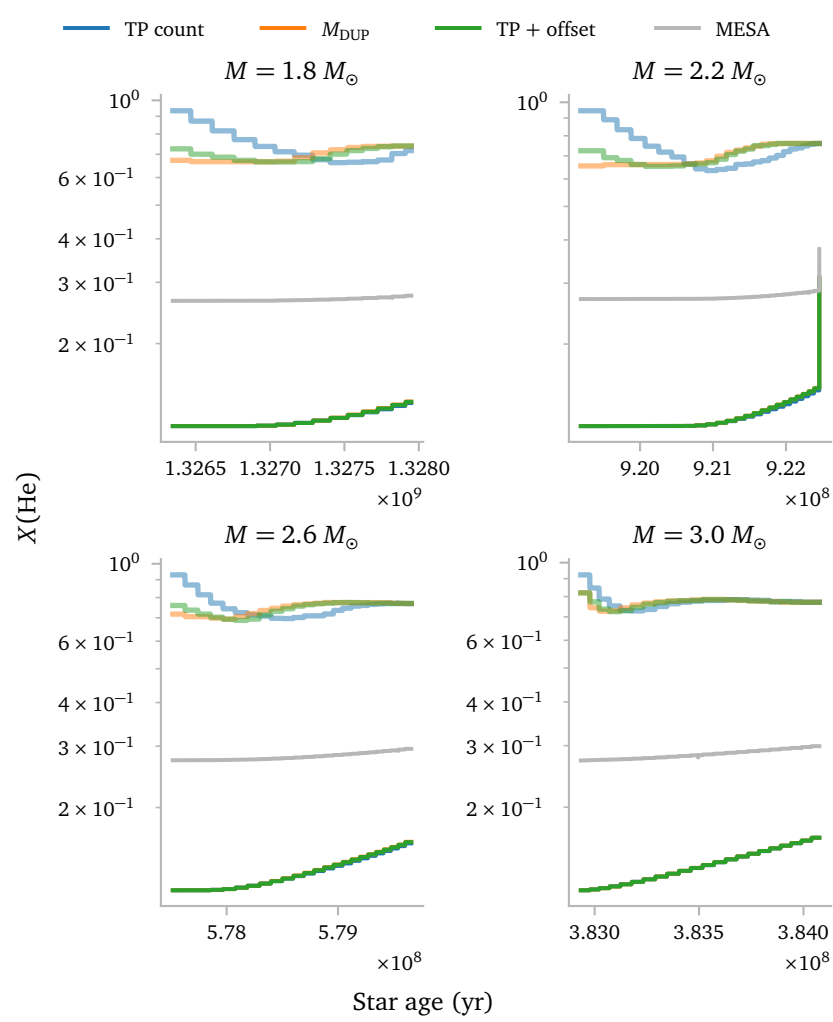
Notes Week 28

a. Monash Initial Helium Envelope Abundances

Last week, I showed these two plots:

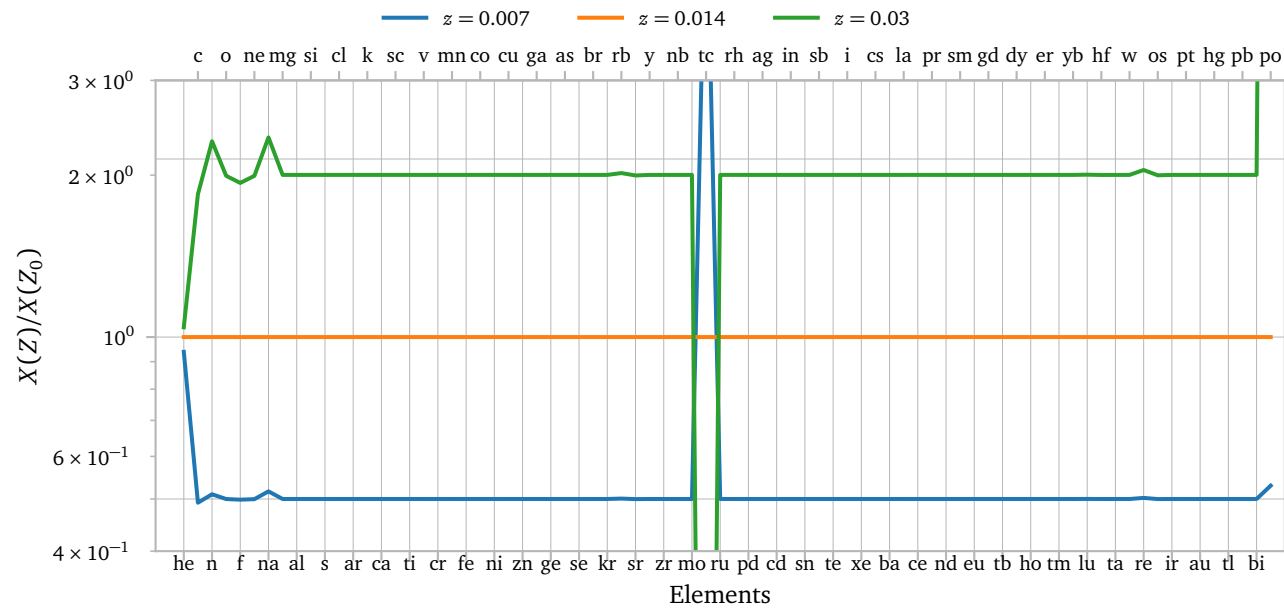


The most important thing we saw here was the quite large differences between the helium mass fractions.



Here, the same difference is again very clear.

It is actually very easy to understand where this discrepancy comes from, and in fact, I have already made a figure before that shows exactly what is going wrong here:

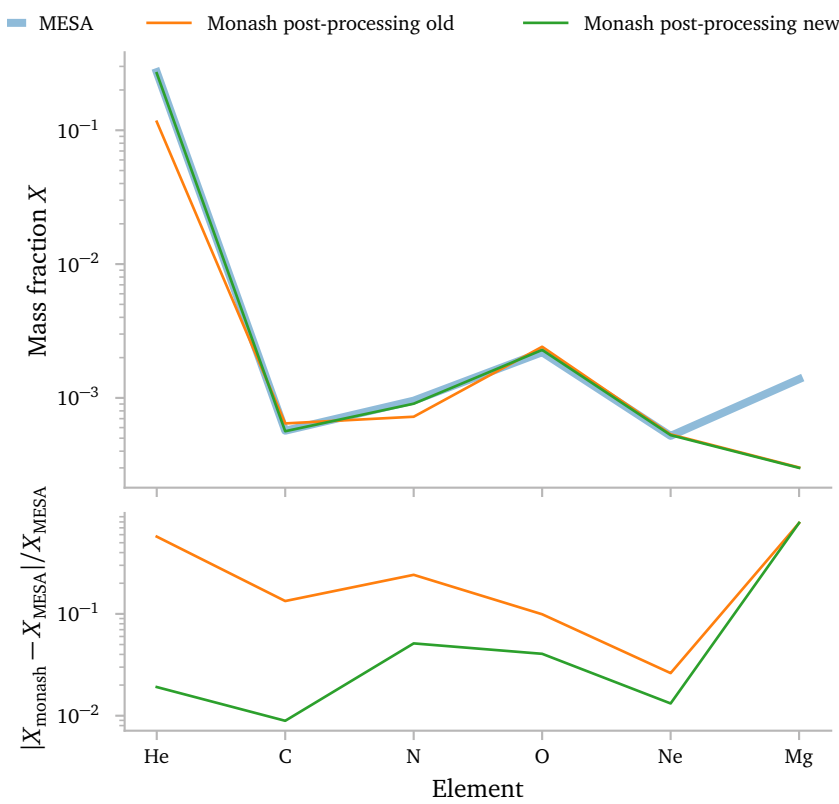


Here I plot the initial envelope abundances for 3 different Monash metallicities divided by the Monash solar Metallicity model.

I made this plot to *confirm* I can simply take the abundance of an element, divide it by the Monash metallicity and multiply it by the MESA metallicity to get the MESA envelope abundance.

Now I think this works great for pretty much all elements, *except* for some of the very light elements, exactly the elements that I was comparing before.

I quickly fixed this by also interpolating the initial envelope abundances.



Clearly, this works much better for helium and actually all other isotopes too.

b. MESA intershell abundances

b.1 Method

b.2 Results

c. Investigating Monash Yields

d. Envelope abundances

References